

Lab 09 JavaFX and collections *(SUMBIT ALL YOUR CODE ON MOODLE)*

Problem 01) Visualizing a Queue with JavaFX

Use JavaFX to create a graphical simulation of a FIFO (First-In-First-Out) queue and LIFO() stack collections. Students will:

- Implement a basic queue using a Collection.
- Create a JavaFX GUI to enqueue and dequeue items.
- Visually represent the queue elements as boxes lined up horizontally.
- Display each queue item as a colored rectangle or label.
- New items appear at the end.
- When dequeued, the first item should disappear.
- Show a message when the queue is empty.

Also use the given CSS styling for a clean, modern look:

- **Styled buttons, text fields, and labels**
- **Colored queue items with rounded borders**
- **Hover effects**

Put the CSS file under the resources folder and then use for the scene as follows:

```
scene.getStylesheets().add(Objects.requireNonNull(getClass().getResource("/queue-style.css")))
```

In the code:

Make a method for adding values to the queue called : `private void enqueue(String value)`

Make a method for removing values to the queue called: `private void dequeue()` (make sure to check if it is empty and print "There's nothing to dequeue!" in that case.

Make a method to clear the scene and redraw the queue called : `private void redrawQueue()`



Problem 02) Visualizing a Stack with JavaFX

Build a JavaFX GUI to illustrate **push** and **pop** operations on a **stack**, with visual elements representing the stack in vertical order.

Stack Concepts Covered

- **Push** adds to the top
- **Pop** removes from the top
- LIFO is visually and behaviorally clear
- Optional: show a “stack underflow” message

Implement the following features:

- Use `Stack<String>` to manage stack data
- Use a **VBox** to visually represent stack items (top to bottom)
- Buttons: Push, Pop, Clear
- Labels: “Top” of the stack

Also use the given CSS styling for a clean, modern look:

- Styled buttons, text fields, and labels
- Colored queue items with rounded borders
- Hover effects

Stack Visualizer

Enter value

Push

Pop

Clear

← Top

Bob

9

Alex

Zak

Chad